

Contamination Signals, Not Verdicts: A Pre-Registered Multi-Probe Audit of Turkish LLM Benchmarks with a Threshold-Sensitivity Protocol

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Abstract

Benchmark contamination—the leakage of evaluation items into training data—threatens the validity of leaderboard rankings and downstream model selection. While contamination auditing has become standard for English benchmarks, non-English evaluation suites remain largely unaudited. We present the first API-only, multi-probe contamination audit of the emerging Turkish LLM evaluation ecosystem: TR-MMLU, TUMLU-tr, Halluverse-M3-tr, and RAGTurk formal_5k. Our pre-registered protocol deploys three independent probe modalities—verbatim recall (M1), distractor-flip (M2a), and cross-lingual 3-arm fragility (M2b)—across three frontier models (GPT-5.6-luna, mimo-v2.5, deepseek-v4-flash), producing 13,193 API calls at a total cost of \$0.71. A threshold-sensitivity analysis distinguishes RAW verdicts (all modalities pooled) from M1-DISCOUNT verdicts (verbatim-recall struck as instruction-following-confounded). Under the adjudicated M1-DISCOUNT rule, **zero of twelve benchmark×model cells are flagged for contamination**. One cell (TUMLU-tr × deepseek-v4-flash) reaches significance under RAW but collapses under M1-DISCOUNT, demonstrating that single-modality signals can mislead without sensitivity analysis. We release all data, code, and pre-registration artifacts for reproducibility.

1 Introduction

The trustworthiness of LLM leaderboards depends on a simple assumption: benchmark items remain unseen by the models being evaluated. When this assumption is violated—through training-data contamination—reported performance reflects memorization rather than capability, undermining model selection and scientific inference (Yang et al., 2024; Liang et al., 2024b; ConStat, 2024).

Contamination detection has become a major research thrust for English benchmarks (Geng et al., 2022; Mattern et al., 2023; Liang et al., 2024a). Yet the non-English evaluation ecosystem is growing faster than the auditing infrastructure that could validate it: new benchmarks for Turkish (Şimşek, Cihan and others; Barov et al., 2025), Hindi (Aggarwal et al., 2024), Bangla (Ahmed et al., 2024), and dozens of other languages appear without corresponding contamination audits.

Turkish presents a particularly acute case. TR-MMLU (Şimşek, Cihan and others) has been publicly available since December 2024—approximately 19 months of exposure to training-data collection pipelines—and a benchmark-quality audit already finds that “70% of the benchmark datasets fail to meet our heuristic quality standards” (TR-MMLU Quality Audit, 2025). The 2025–2026 Turkish evaluation wave (TUMLU (Barov et al., 2025), RAGTurk (METU NLP, 2026), Halluverse-M3 (Dalğar et al., 2026)) adds four new benchmarks to the ecosystem, none audited for contamination.

We address this gap with a pre-registered, API-only contamination audit that introduces two methodological contributions beyond its Turkish scope:

1. **A threshold-sensitivity protocol** that distinguishes verdicts under pooled modalities (RAW) from verdicts after striking the instruction-following-confounded M1 modality (M1-DISCOUNT), preventing false positives from the most common contamination signal.
2. **A multi-probe consensus architecture** requiring ≥ 2 of 3 independent modalities to agree before flagging a cell, following the finding that single-probe methods degrade at scale (et al., 2024).

Our audit covers four Turkish benchmarks × three frontier models × three probe types, producing 13,193 API calls at \$0.71 total cost—demonstrating that rigorous contamination auditing is accessible without large compute budgets.

2 Related Work

Benchmark contamination detection. The contamination detection literature has evolved from simple n-gram overlap (Brown et al., 2020) to membership-inference attacks (Geng et al., 2022), perplexity-based methods (Oren et al., 2023), and temporal-signal analysis (Zhang et al., 2025). Carlini et al. (Carlini et al., 2021) demonstrate training-data extraction feasibility, establishing the threat model that contamination audits operationalize. Yang et al. (Yang et al., 2024) provide a taxonomy of contamination types—verbatim memorization, paraphrase contamination, and structural leakage—that our three-probe design directly maps onto. Recent work has moved toward multi-signal approaches: the Contamination Multiplier framework (Contamination Multiplier, 2024) combines multiple detection signals with calibrated thresholds. ConStat (ConStat, 2024) operationalizes contamination as “non-generalizing performance,” providing the theoretical foundation for our M2b cross-lingual fragility probe. Sánchez Salido et al. (Sánchez Salido et al., 2025) introduced MCQ variation separating memorization from reasoning—our M2a probe extends this principle using distractor-flip editing.

Non-English benchmark auditing. The Bangla MathShikkha audit (Ahmed et al., 2024) establishes that contamination auditing is accepted for non-English benchmarks. Beyond Benchmarks (Beyond Benchmarks, 2025) surveys evaluation threats across languages. LINGOly (LINGOly, 2024) audits multilingual LLMs across 44 languages. For Turkish, TR-MMLU (Şimşek, Cihan and others) did not include contamination auditing; the quality audit (TR-MMLU Quality Audit, 2025) documented structural weaknesses but not leakage. Yild

iz et al. (Yıldız et al., 2025) evaluated quality across 17 Turkish benchmarks. Abbas et al. (Abbas et al., 2026) showed translation suppresses contamination indicators in Arabic—our M2b probe’s 3-arm design (TR → back-translation → EN direct) extends this insight to Turkish.

Pre-registration and honest reporting. Pre-registration—fixing analysis rules before seeing results—is standard in clinical trials and increasingly advocated for ML evaluation (Pineau et al., 2020). The ML Reproducibility Crisis (Gencoglu et al., 2019) documents how post-hoc analysis choices inflate results. Our protocol pre-registers the exact decision rule, sample size, seed, and consensus architecture before any API calls. The “Contamination Signals, Not Verdicts” framing follows the growing recognition that contamination auditing produces graded signals, not binary classifications (GAOKAO-Eval, 2024).

3 Method

3.1 Benchmarks and Models

We audit four Turkish LLM evaluation benchmarks (Table 1) across three frontier models tested via the opencode-go API (Table 2).

3.2 Probe Design

Three independent probe modalities target different contamination signatures:

M1—Verbatim recall. Question and answer options presented with position labels stripped; the model completes the item freely. We measure 8-gram overlap and correct-option verbatim rate. M1 is confounded by instruction-following (models produce correct answers regardless of contamination), so we strike it in the M1-DISCOUNT reading.

M2a—Distractor flip. One distractor is replaced with a correct answer from a different item. If the model follows the planted answer, it indicates surface-level memorization rather than conceptual understanding. Significance tested with McNemar’s exact test.

M2b—Cross-lingual 3-arm fragility. Three conditions per item: (A) original Turkish, (B) back-translated Turkish (TR→EN→TR), (C) English-direct. The contamination signature is **B collapses while C stays stable**—the model is anchored to the memorized Turkish surface rather than the underlying concept (ConStat, 2024). Significance tested with McNemar’s exact test; Bonferroni-corrected per benchmark × probe family. For RAGTurk (free-form QA), M2a is not applicable.

3.3 Consensus and Adjudication

A cell is flagged **contaminated** only if: (1) ≥ 2 of 3 modalities independently indicate contamination, AND (2) ≥ 1 dynamic arm has $p < 0.05$ after Bonferroni correction. We report two readings per cell:

- **RAW:** All three modalities pooled.
- **M1-DISCOUNT:** M1 struck; M2a and M2b must indicate independently.

M1-DISCOUNT is the **adjudicated** call. Cells flagged by RAW but not M1-DISCOUNT are labeled **THRESHOLD-DEPENDENT**—never “contaminated.” All rules, $n = 200$, seed 42, and cost cap \$2.00 were pre-registered before any API calls.

4 Results

Run reliability. The audit completed with 13,193 logged API calls (13,154 succeeded, 99.7%; 39 failed; 255 retries). Total cost: \$0.71 against a \$2.00 cap.

Benchmark	Source	Format	Items (sampled)	Public since
TR-MMLU	Şimşek (Şimşek, Cihan and others)	MC-QA, 62 sections	200	2024-12-31
TUMLU-tr	Barov et al. (Barov et al., 2025)	MC-QA, 8 Turkic languages	200	2025-02-16
Halluverse-M3-tr	Dalğar et al. (Dalğar et al., 2026)	QA + dialogue summarization	200	2026-02-06
RAGTurk formal_5k	METU NLP (METU NLP, 2026)	QA + passage chunks	200	2026-02-03

Table 1: Turkish LLM evaluation benchmarks audited. All items are seed-42 matched across probes.

Model	Provider	Cost (rated)	Notes
GPT-5.6-luna	OpenAI	\$0.348	Subject model
mimo-v2.5	Xiaomi	\$0.000 (uncosted)	Subject model
deepseek-v4-flash	DeepSeek	\$0.358	Reference cohort

Table 2: Models tested in the contamination audit.

4.1 MCQ Benchmarks: TR-MMLU and TUMLU-tr

Table 3 presents combined results for the two MCQ benchmarks.

TR-MMLU: M1 rates are high for instruction-following models (expected, M1 confounded). M2a flip rates are low (9–16%). M2b B-vs-A is significant for GPT-5.6-luna and deepseek, but C-vs-A is not for any model—the pattern is back-translation noise, not surface-anchored memorization. **Adjudicated: No contamination detected** (all models, both readings).

TUMLU-tr: Strongest M2b B-vs-A signals—all significant after Bonferroni. For GPT-5.6-luna and deepseek, C-vs-A is also significant/near-significant, suggesting general translation quality degradation. The critical cell: **deepseek-v4-flash** × **TUMLU-tr** reaches significance under RAW (M1 rate 0.570 + M2b B-vs-A $p < 0.001 = 2/3$ modalities). Under M1-DISCOUNT, only M2b indicates (1/3), failing the threshold. **Adjudicated: THRESHOLD-DEPENDENT (M1-sensitive)** for deepseek-v4-flash; no contamination for GPT-5.6-luna and mimo-v2.5.

4.2 Free-Form QA: Halluverse-M3-tr and RAGTurk

Table 4 presents the two free-form QA benchmarks.

Halluverse-M3-tr: Zero verbatim recall across all models (clean negative M1 signal). M2a flip rates low (9–11%). M2b baseline is low (2.5–5.5% Arm A), limiting fragility contrast. **Adjudicated: No contamination detected.**

RAGTurk formal_5k: Accuracies are lower bounds under the precision-first matcher (recall 0.371 on 106 labeled pairs). Arm C zero accuracy reflects English-translated QA scoring difficulty, not model failure. **Adjudicated: No contamination detected.**

4.3 Adjudication Summary

Table 5 presents the full adjudication across all 12 cells.

Headline: 0/12 cells flagged under the pre-registered M1-DISCOUNT rule. 1/12 cells flagged under RAW but reclassified as THRESHOLD-DEPENDENT.

5 Discussion

The M1-DISCOUNT distinction matters. Without sensitivity analysis, deepseek-v4-flash × TUMLU-tr would be reported as “contaminated” based on RAW modalities. The M1 verbatim rate of 0.570—which looks like a contamination signal—is instruction-following behavior: when asked to “answer the question,” models produce correct answers regardless of memorization. Striking M1 reveals only M2b indicates (1/3 modalities), failing the threshold. This generalizes beyond Turkish: any audit pooling M1-style signals with dynamic arms risks false positives without sensitivity analysis. We recommend the RAW + M1-DISCOUNT dual-reporting convention.

Honest negatives are scientific contributions. All 12 cells produce honest negatives under the adjudicated rule: four Turkish benchmarks, three frontier models, 13,193 API calls, no contamination detected. Null findings are citable contributions when the protocol is pre-registered and the sample is adequate (Pineau et al., 2020).

Cost accessibility. The \$0.71 total cost demonstrates that rigorous contamination auditing is accessible to independent researchers without institutional compute budgets.

Limitations. (1) Black-box API: training-data inclusion is unobservable; weight-based methods would provide stronger evidence but require model access. (2) M1 confound: handled via M1-DISCOUNT but RAW is reported for transparency. (3) QA matching: RAGTurk matcher is precision-first (1.000 precision, 0.371 recall); all RAGTurk accuracies are conservative lower bounds. (4) Sample size: $n = 200$ is a sample, not the full benchmark. (5) Translation noise: RAGTurk Arm C golds are machine-translated. (6) Scope: findings

Benchmark	Model	M1 verbatim rate	M2a flip rate	M2b B-vs-A p	M2b C-vs-A p
TR-MMLU	GPT-5.6-luna	0.860 [0.805, 0.901]	0.155 [0.111, 0.212]	<0.001	1.000
	mimo-v2.5	0.575 [0.506, 0.641]	0.140 [0.099, 0.195]	0.150	1.000
	deepseek-v4-flash	0.780 [0.718, 0.832]	0.145 [0.103, 0.201]	0.004	0.239
TUMLU-tr	GPT-5.6-luna	0.885 [0.833, 0.922]	0.096 [0.062, 0.144]	<0.001	0.040
	mimo-v2.5	0.365 [0.301, 0.434]	0.111 [0.074, 0.162]	<0.001	1.000
	deepseek-v4-flash	0.570 [0.501, 0.637]	0.111 [0.074, 0.162]	<0.001	0.831

Table 3: MCQ benchmark probe results. TR-MMLU: M2b B-vs-A is significant for GPT-5.6-luna and deepseek, but C-vs-A is not—pattern is back-translation noise, not surface-anchored memorization. TUMLU-tr: All models show significant M2b B-vs-A; for GPT-5.6-luna and deepseek, C-vs-A is also significant/near-significant, suggesting general translation degradation.

Benchmark	Model	M1 verbatim rate	M2a flip rate	M2b Arm A acc.	M2b Arm C acc.
Halluverse-M3-tr	GPT-5.6-luna	0.000 [0.000, 0.019]	0.090 [0.058, 0.138]	0.025	—
	mimo-v2.5	0.000 [0.000, 0.019]	0.085 [0.054, 0.132]	0.045	—
	deepseek-v4-flash	0.000 [0.000, 0.019]	0.110 [0.074, 0.161]	0.055	—
RAGTurk formal_5k	GPT-5.6-luna	0.052 [0.028, 0.093]	—	0.080	0.000
	mimo-v2.5	0.030 [0.014, 0.064]	—	0.025	0.000
	deepseek-v4-flash	0.070 [0.042, 0.114]	—	0.070	0.000

Table 4: Free-form QA benchmark probe results. M2a is not applicable for RAGTurk (no MCQ structure). Arm C for RAGTurk is scored against machine-translated golds. Halluverse-M3-tr: — for Arm C indicates QA-specific scoring.

apply only to these four benchmarks under these three models.

6 Conclusion

We present the first multi-probe contamination audit of the Turkish LLM evaluation ecosystem. Under M1-DISCOUNT, no contamination is detected in any of the 12 cells. The RAW + M1-DISCOUNT dual-reporting convention generalizes beyond Turkish to any contamination audit pooling verbatim-recall signals with dynamic-arm signals. We release all data, code, and pre-registration artifacts for reproducibility.

Appendix: Spend Log

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Benchmark	Model	RAW	M1-DISCOUNT
TR-MMLU	GPT-5.6-luna	—	—
TR-MMLU	mimo-v2.5	—	—
TR-MMLU	deepseek-v4-flash	—	—
TUMLU-tr	GPT-5.6-luna	—	—
TUMLU-tr	mimo-v2.5	—	—
TUMLU-tr	deepseek-v4-flash	Flagged (2/3)	Flagged (1/3)
Halluverse-M3-tr	GPT-5.6-luna	—	—
Halluverse-M3-tr	mimo-v2.5	—	—
Halluverse-M3-tr	deepseek-v4-flash	—	—
RAGTurk formal_5k	GPT-5.6-luna	—	—
RAGTurk formal_5k	mimo-v2.5	—	—
RAGTurk formal_5k	deepseek-v4-flash	—	—

Table 5: Adjudication summary. — = not flagged. Under M1-DISCOUNT, **0/12 cells flagged**; under RAW, **1/12 threshold-dependent**.

Model	Calls	OK	Fail	Retries	Cost
GPT-5.6-luna	3,798	3,759	39	255	\$0.348
mimo-v2.5	3,798	3,798	0	0	\$0.000
DeepSeek-v4-flash	5,597	5,597	0	0	\$0.358
TOTAL	13,193	13,154	39	255	\$0.707

Table 6: Spend log summary. Total project spend including preparatory runs: \$0.89.

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